NEST360 Implementation Model

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Why are we doing this?

NEST360 Impact:
Mortality Interim Results for Africa's Newborns



NEST360 implementation of neonatal care across Africa

NEST360, an alliance of 23 organizations, has partnered with governments across 5 African countries to deliver lifesaving care for vulnerable newborns in 65 hospitals with phased starts since 2019. The NEST360 implementation approach is based on an individual-patient level dataset that includes more than 500,000 newborn admissions so far.

NEST360 enables sustainable care for vulnerable babies with a systems approach focused on:

- **lifesaving newborn care technologies**
- **education ecosystem** for healthcare workers and biomedical engineers
- **data-driven action** for improving quality newborn care based on local government data

Saving lives is driven by coverage of high-impact care

NEST360 found that mortality change signals emerged in 2023 with improved coverage, coinciding with data use and quality improvement strategies focused on coverage and quality of care, based on our interrupted time series analysis. We also found that following the installation of the NEST360 bundle of devices and training, there was a lack of a step change in mortality.

Based on these findings, NEST360 collaboratively set hospital-specific coverage targets to accelerate measurable impact for key interventions, notably breathing support for preterm babies (CPAP), Kangaroo Mother Care (KMC), phototherapy use, and preventing hypothermia. We aim to achieve the targets in these 65 hospitals by the end of 2025.

EXCITING MORTALITY CHANGE THAT COUNTS!

Interim findings comparing 2024 with 2021/2022 (full results due in 2026)

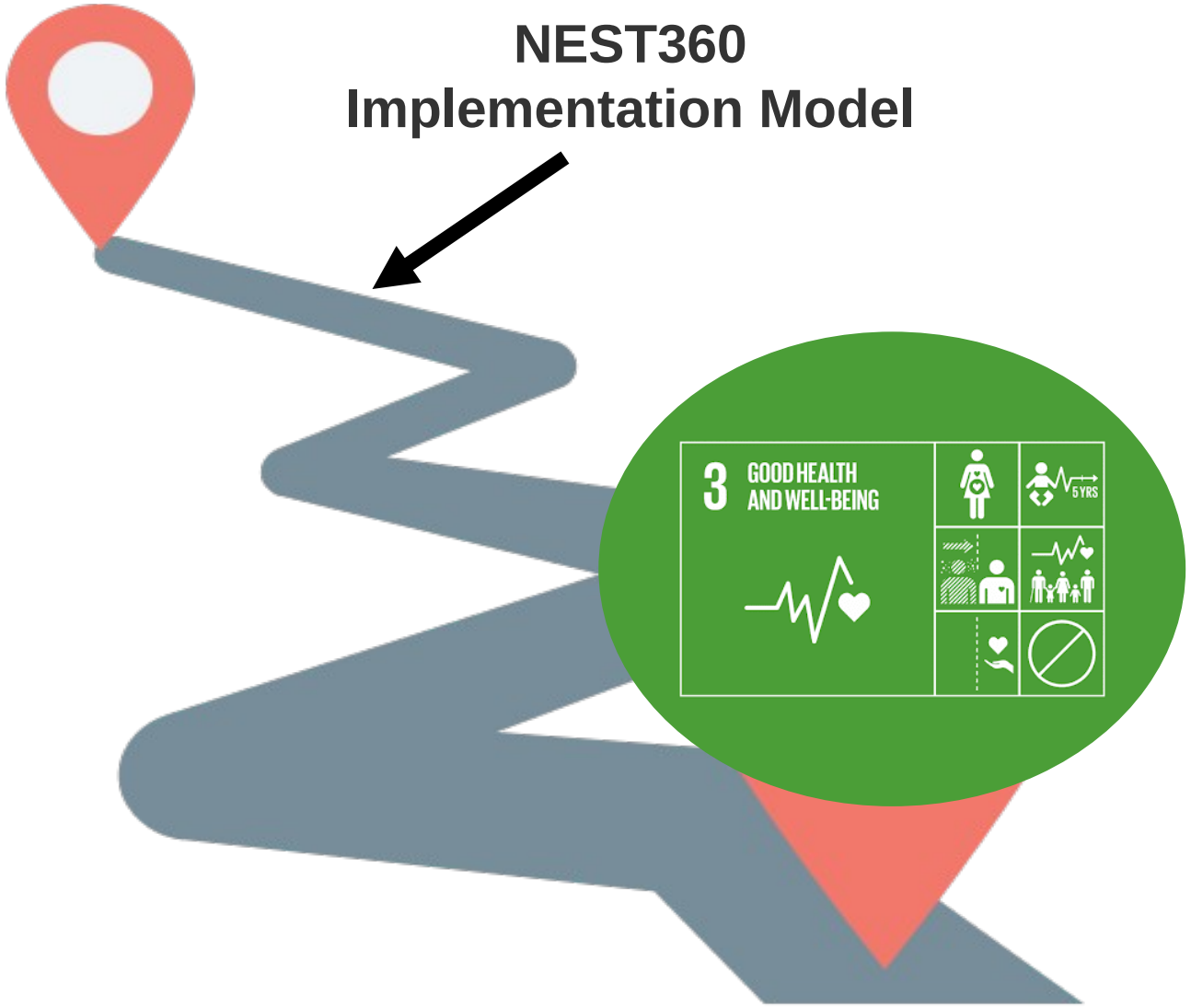
- Significant mortality reduction for admitted newborns across all four countries, **particularly for those weighing 1000–1999g and >2000g**
- Mortality change varied between hospitals, **with significant mortality reduction ranging from 14–54%**
- **All 4 countries have hospitals among the top fast progressors**, with significant increases in high-impact interventions, particularly CPAP and KMC
- All countries have one or more hospitals with an apparent increase in mortality. **Many hospitals have system challenges** such as small spaces with overcrowding, lack of nurses, and/or power outages
- Notably, **Malawi has achieved a 16% national-level reduction in neonatal mortality in hospitals as of 2024**



OUR REACH ACROSS AFRICA

5 countries | 144 hospitals

NEST360 Implementation Model



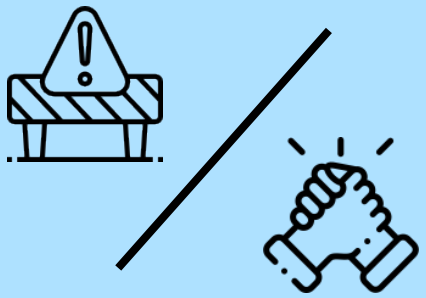
What is an implementation model?

A visual tool that helps us to systematically plan, report, and synthesize how NEST360 implemented small and sick newborn care

The core elements of the model are implementation factors, strategies, mechanisms of action, and outcomes

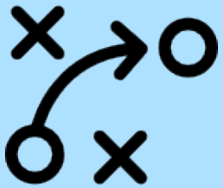
What are the components of the model?

Implementation factors



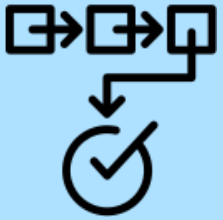
The barriers and facilitators that influence the success of implementing SSNC in level 2+ neonatal units

Strategies



The activities NEST used to overcome barriers, leverage facilitators, and implement SSNC in our supported countries

Mechanisms of Action



The processes or pathways through which a strategy produces change in outcomes

Outcomes



The indicator used to assess how well an intervention is implemented into practice

How is a NEST360 implementation model useful?



Tells the full story of our implementation work



Enhance communication with funders and partners

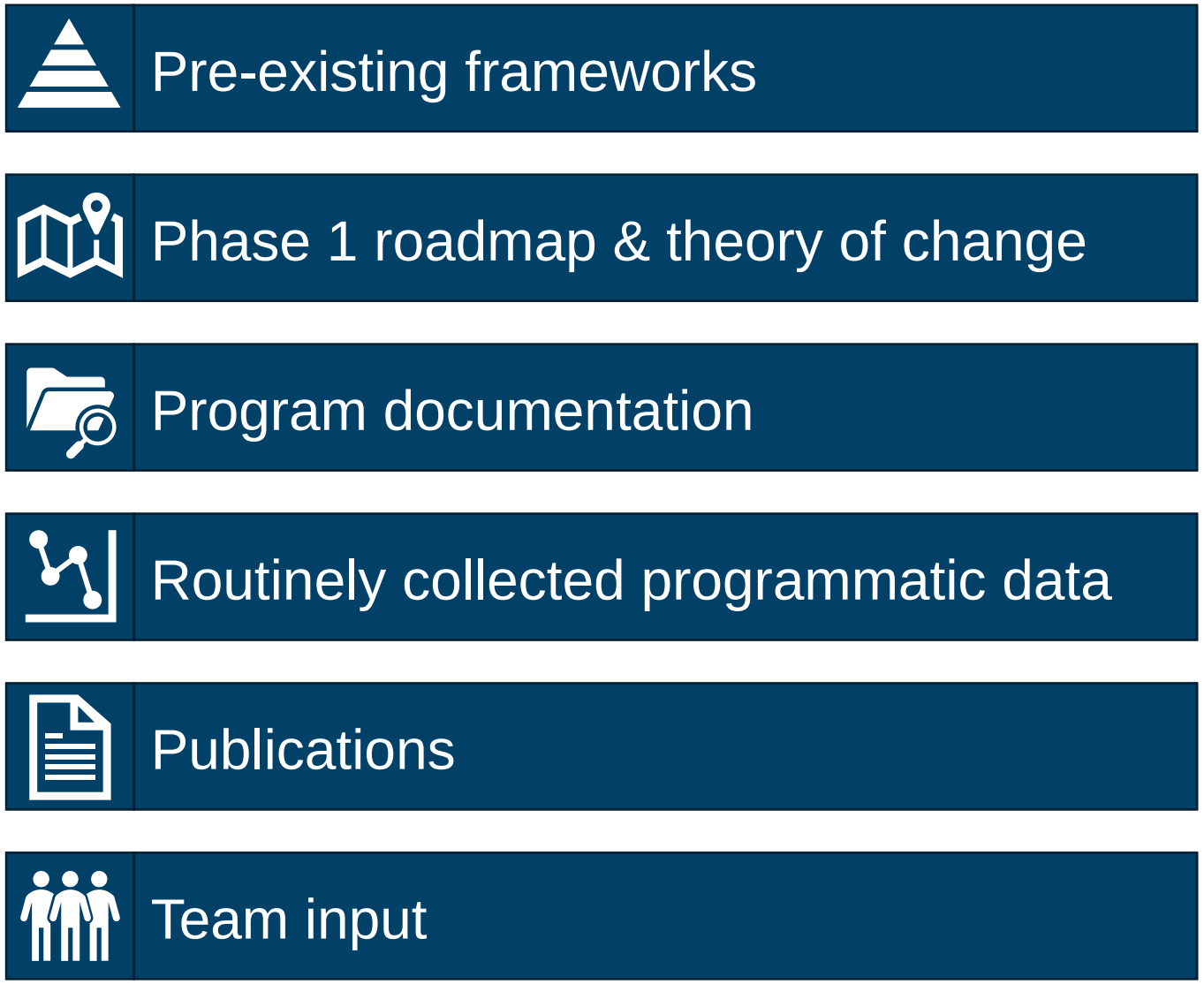


Supports planning, evaluation, and adaptations



Supports long-term sustainability and scale-up decisions for Phase 2

Data sources to develop the NEST360 implementation model



The NEST360 implementation model is a multi-tiered guide that…

Reports on NEST360's work

- Describes the implementation environment where NEST360 operates
- Details the strategies and activities used to reduce newborn mortality
- Presents outcomes tracked by NEST360 along with the overall program impact

Provides a model others can adapt to implement similar efforts

- Identified key implementation factors to consider during planning
- Connects to existing NEST360 tools and resources that support implementation
- Defines essential processes and impact measures to monitor throughout implementation

Tiers within the model

Tier 1

Provides a graphic that can be added to grant proposals or shared with funders and stakeholders

Tier 2

Provides general information that other implementers should consider if they'd like to implement a "NEST-like model"

Tier 3

Provides the details one could need to gain a comprehensive understanding of how NEST360 implemented our package

NEST360 Implementation Model: Tier 1

Implementation Factors



SSNC core components/ systems components

Intervention: Evidence-based small and sick newborn care in Level 2+ units

Strategies used by NEST360

Life saving technology

Education ecosystem

Data-driven action

Multi-level sustainable systems change

Mechanisms

How strategies interact with implementation factors

Outcomes

Global

National/Country

Hospital

Impact

Reduction in newborn mortality!

Tier 2

This is where we provide details related to implementing the NEST model, but it remains neutral in stance

Intervention

Evidence-based Sick and Small Newborn Care in Level 2

An expanded list of implementing factors organized by SSNC implementing factors

Implementation Factors (SSNC Core Components)

Leadership & Governance
- National plans and policies for SSNC
- Facility level guidelines and protocols for SSNC
- Engagement with MOH
- Engagement with facility-level leadership
- Facility champions
- Global and national focus on SSNC and NMR
- Integration of SSNC into current policy & funding

Data Systems
- Availability of neonatal health forms, registers, and data systems at facility level capturing SSNC sensitive indicators
- SSNC sensitive indicators at facility and national levels
- Data access and utilization
- Data quality
- Data management and governance
- d=Dedicated neonatal data clerk

Human Resources
- Health workers attitudes/interest in SSNC
- Health workers competence/confidence in SSNC
- Preservice training in SSNC
- Training available for SSNC
- BMET training
- HC Provider awareness /knowledge
- Supporting staff following IPC guidelines (janitors, guard, support runners)
- Available BMETs
- Adequate number of health workers
- Little to no staff rotation or turnover

Quality Improvement
- Consistent QI knowledge at national, regional, and facility levels
- Active hospital and ward improvement teams
- Consistent QI training
- Consistent availability of QI documentation tools
- Documentation completeness
- Consistant auditing of care quality (ex. MPDSA)
- National and regional QI structures and resources include SSNC-focused guidance

Equipment & Commodities
- Availability of functional equipment and supplies at facilities
- Strong procurement system
- Technology fit for purpose
- Skill labs/skill stations available and used
- Affordable technology
- No or infrequent stockouts
- No or infrequent power outages

Referral process
- Adherence to referral guidelines
- Strong pre and during referral communication and consultations
- Strong referral feedback mechanism
- Appropriate transportation for referrals

Linkage to Maternal Care
- Policies that facilitate/enable zero separate
- Integrated antenatal, delivery, postnatal, and SSNC
- Women staying in the hospital with their newborns

Family centered care
- Training and policies to support parental involvement
- Education/counseling for caregivers
- Family members are aware of handwashing and IPC practices
- Family interest in being involved in the care process
- Family involvement in decision making process
- Availability of bereavement care

Post-discharge follow up
- Discharge follow up and early child development
- Follow up feedback mechanisms
- Community health workers engagement
- Community based support and continuous care

Infrastructure
- Standard guide for space for Level 2 SSNC
- Adequate space for SSNC including KMC/immediate KMC
- Appropriately designed space - leads to easy flow and strong IPC practices
- Flow of care of newborns from labor ward to NBU
- Reliable electric power and poor backup power sources
- Medical gas and vacuum systems
- Sufficient WASH facilities
- Blood culture are available

Financing
- Donor funding
- Government financing
- Long-term funding
- Budget allocations for SSNC in national and facility budgets
- No poverty
- Insurance coverage
- Exempted service reimbursement system

Implementation Strategies

1.Engage policy makers & leadership

2.Quality improvement (QI) activities

6.Interprofessional collaboration & teamwork

7.Motivate staff and empower collaboration

8.Ensuring the availability of appropriate functional SSNC medical devices

9.Identify & qualify appropriate technology

10.Hatch Technologies (Africa-based technology distirbution)

11.Engage family members in a care process and follow up actions

12.Ensure adequate funding

13.Tailoring local infrastructure

Mechanisms

1. Policy-maker and leadership engagement fosters alignment of SSNC with national health priorities, **promotes long-term commitment,** and ensures **integration of SSNC into broader health system goals**

2.Routine data-driven QI cycles build a **culture of learning and accountability,** enabling timely identification of gaps and driving sustainable quality

ems enable urse correction,

ucational , improves **adoption and adherence to**

s provider bility, **improving** dge, and **ability of SSNC**

6.Strengthening collaboration across cadres enhances communication, shared ownership, and problem solving, promoting **fidelity to standards of care, increasing adoption of EBIs across cadres,** and **improving continuity of SSNC delivery**

7.Recognition, supportive supervision, and empowerment of staff **increases acceptability, motivation, and engagement** leading to greater accountability, reduced burnout, and **higher qua care delivery**

8.Ensuring appropriate, functional technology i available at the right level of care enables provider delivery life-saving interventions safely and effective promoting **feasibility and adoption of EBIs,** and **reducing gaps in SSNC delivery**

9.Identifying, testing, and sharing results of devices t can function in rugged settings **increases the feasibi of their use to provide SSNC preventing device graveyards** and **increasing the adoption of SSNC EBI** because devices remain functional and available

10.Reliable, context-appropriate technology distr channels improve **equitable access to devices, re downtime** from broken or missing equipment, and promotes **sustainability of SSNC infrastructure in low-resource settings**

11.Actively involving families in care builds trust, **increases caregiver competence and adherence to recommended practices,** and **promotes continuity of care after discharge,** improving neonatal survival

12.Securing and sustaining funding streams enables consistent procurement of supplies, **maintenance of infrastructure,** and **retention of staff capacity,** ensuring **sustainability of SSNC services**

13.Tailoring local infrastructure creates **safer, more efficient care environments, improving feasibility, quality,** and **sustainability of SSNC services**

Outcomes

Leadership and governance
-Global-level
-National-level: SSNC and biomed allocations in annual budget
-Hospital-level:

Data systems
-Global-level: # of MOH NEST-IT logins
-National-level: % of countries logging into NEST-IT monthly
-Hospital-level: Data entry delays and data validity

Human resources
-Global-level: # of clinicians that have receive NEST SSNC training
-National-level: % of hospitals with SSNC trained staff
-Hospital-level: Trained biomed at the hospital (Y/N)

Quality improv
-Global-level
-National-lev
-Hospital

-Global-level
-National-level
-Hospital-level

Financing
-Global-level
-National-level
-Hospital-level

FINAL IMPACT

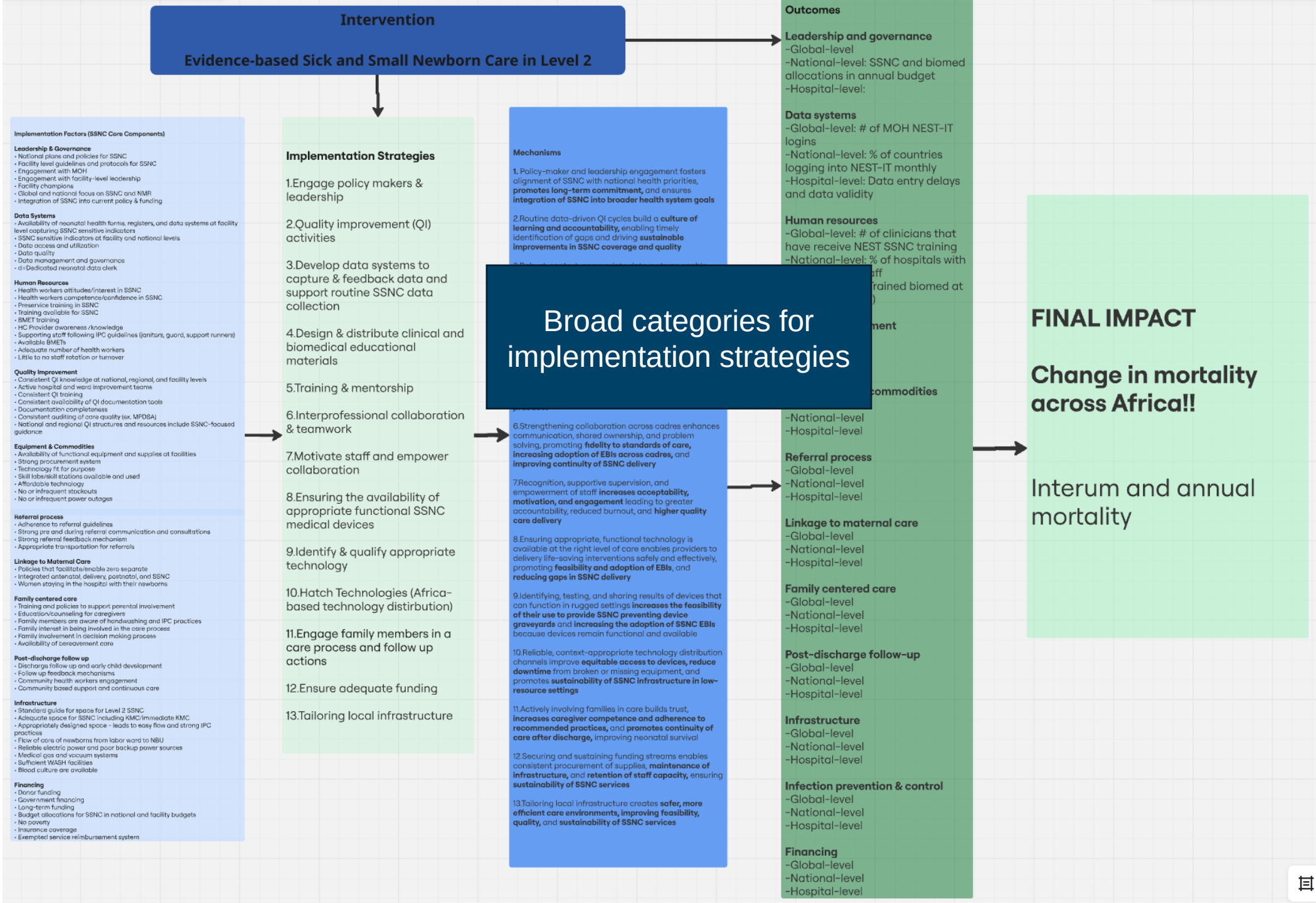
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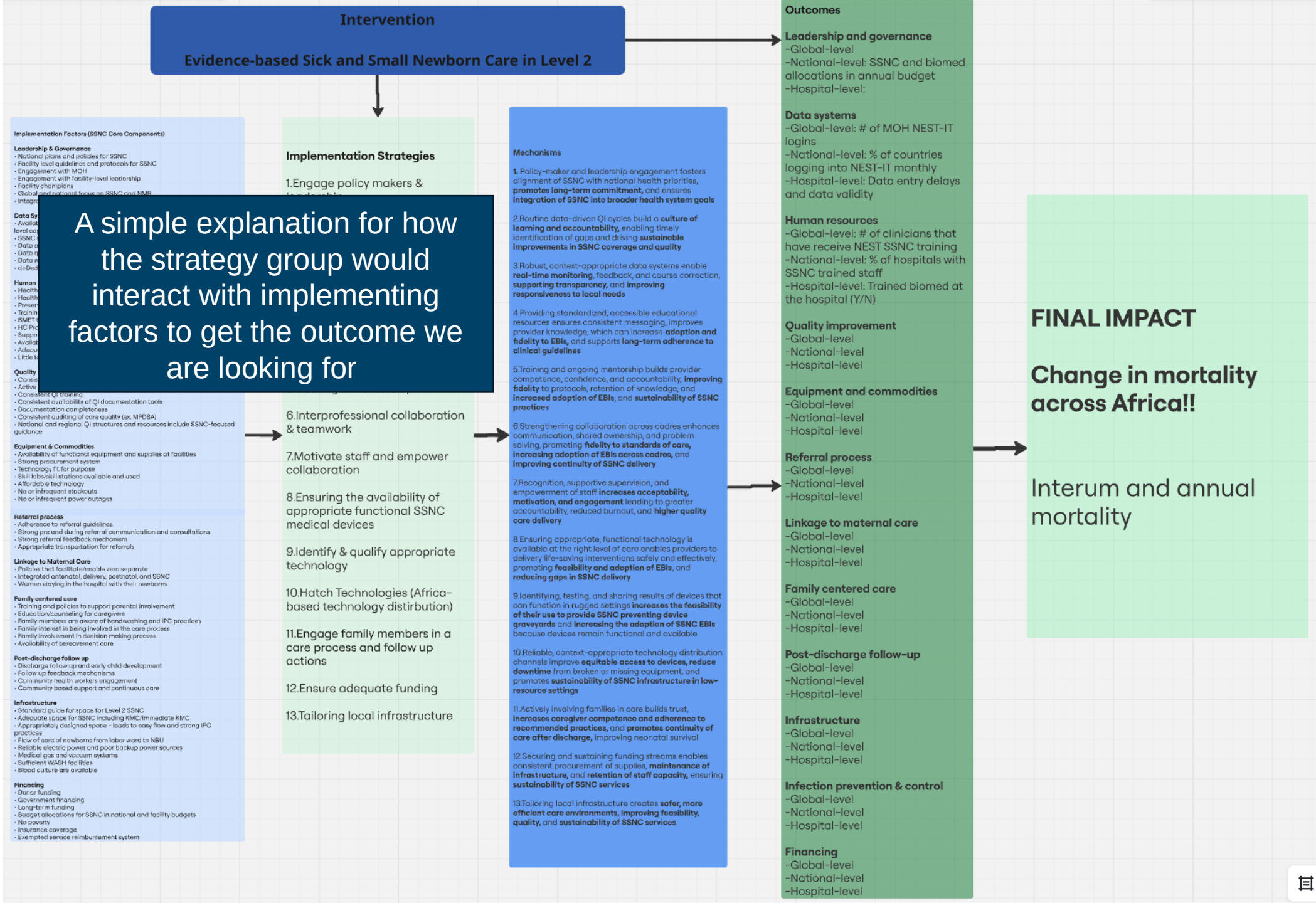
Tier 2

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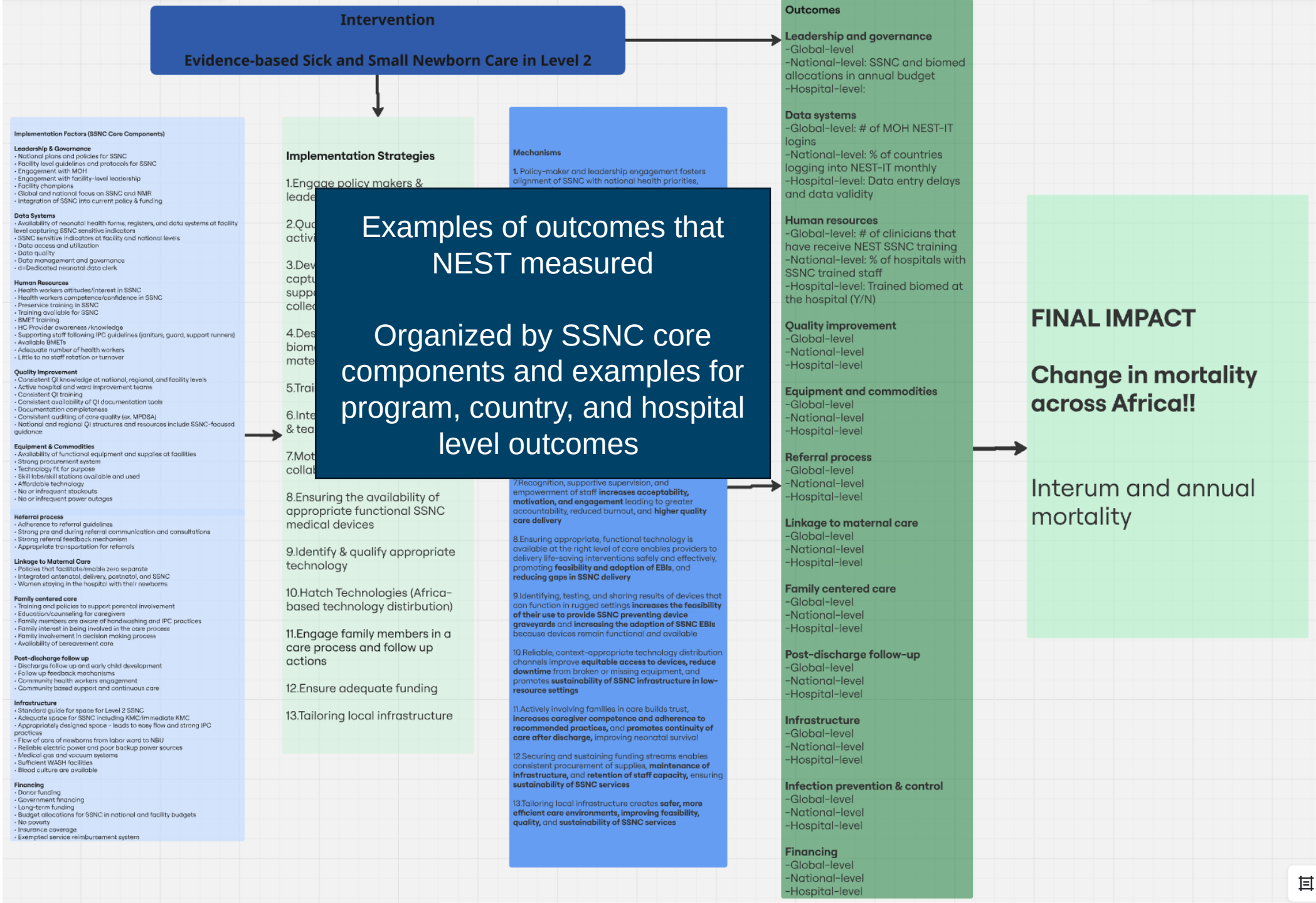
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Tier 2

This is where we provide details related to implementing the NEST model, but it remains neutral in stance



Tier 2: Implementation Factors Part 1

- **Leadership & Governance**
  - National plans and policies for SSNC
  - Facility level guidelines and protocols for SSNC
  - Engagement with MOH
  - Engagement with facility-level leadership
  - Facility champions
  - Global and national focus on SSNC and NMR
  - Integration of SSNC into current policy & funding
- **Data Systems**
  - Availability of neonatal health forms, registers, and data systems at facility level capturing SSNC sensitive indicators
  - SSNC sensitive indicators at facility and national levels
  - Data access and utilization
  - Data quality
  - Data management and governance
  - Dedicated neonatal data clerk
- **Human Resources**
  - Health workers attitudes/interest in SSNC
  - Health workers competence/confidence in SSNC
  - Preservice training in SSNC
  - Training available for SSNC
  - BMET training
  - HC Provider awareness /knowledge
  - Supporting staff following IPC guidelines (janitors, guard, support runners)
  - Available BMETs
  - Adequate number of health workers
  - Little to no staff rotation or turnover
- **Quality Improvement**
  - Consistent QI knowledge at national, regional, and facility levels
  - Active hospital and ward improvement teams
  - Consistent QI training
  - Consistent availability of QI documentation tools
  - Documentation completeness
  - Consistent auditing of care quality (ex. MPDSA)
  - National and regional QI structures and resources include SSNC-focused guidance
- **Equipment & Commodities**
  - Availability of functional equipment and supplies at facilities
  - Strong procurement system
  - Technology fit for purpose
  - Skill labs/skill stations available and used
  - Affordable technology
  - No or infrequent stockouts
  - No or infrequent power outages
- **Referral process**
  - Adherence to referral guidelines
  - Strong pre and during referral communication and consultations
  - Strong referral feedback mechanism

Tier 2: Implementation Factors Part 2

- **Linkage to Maternal Care**
  - Policies that facilitate/enable zero separate
  - Integrated antenatal, delivery, postnatal, and SSNC
  - Women staying in the hospital with their newborns
- **Family centered care**
  - Training and policies to support parental involvement
  - Education/counseling for caregivers
  - Family members are aware of handwashing and IPC practices
  - Family interest in being involved in the care process
  - Family involvement in decision making process
  - Availability of bereavement care
- **Post-discharge follow up**
  - Discharge follow up and early child development
  - Follow up feedback mechanisms
  - Community health workers engagement
  - Community based support and continuous care

- **Infrastructure**
  - Standard guide for space for Level 2 SSNC
  - Adequate space for SSNC including KMC/immediate KMC
  - Appropriately designed space - leads to easy flow and strong IPC practices
  - Flow of care of newborns from labor ward to NBU
  - Reliable electric power and poor backup power sources
  - Medical gas and vacuum systems
  - Sufficient WASH facilities
  - Blood culture are available
- **Financing**
  - Donor funding
  - Government financing
  - Long-term funding
  - Budget allocations for SSNC in national and facility budgets
  - No poverty
  - Insurance coverage
  - Exempted service reimbursement system

Tier 2 strategies

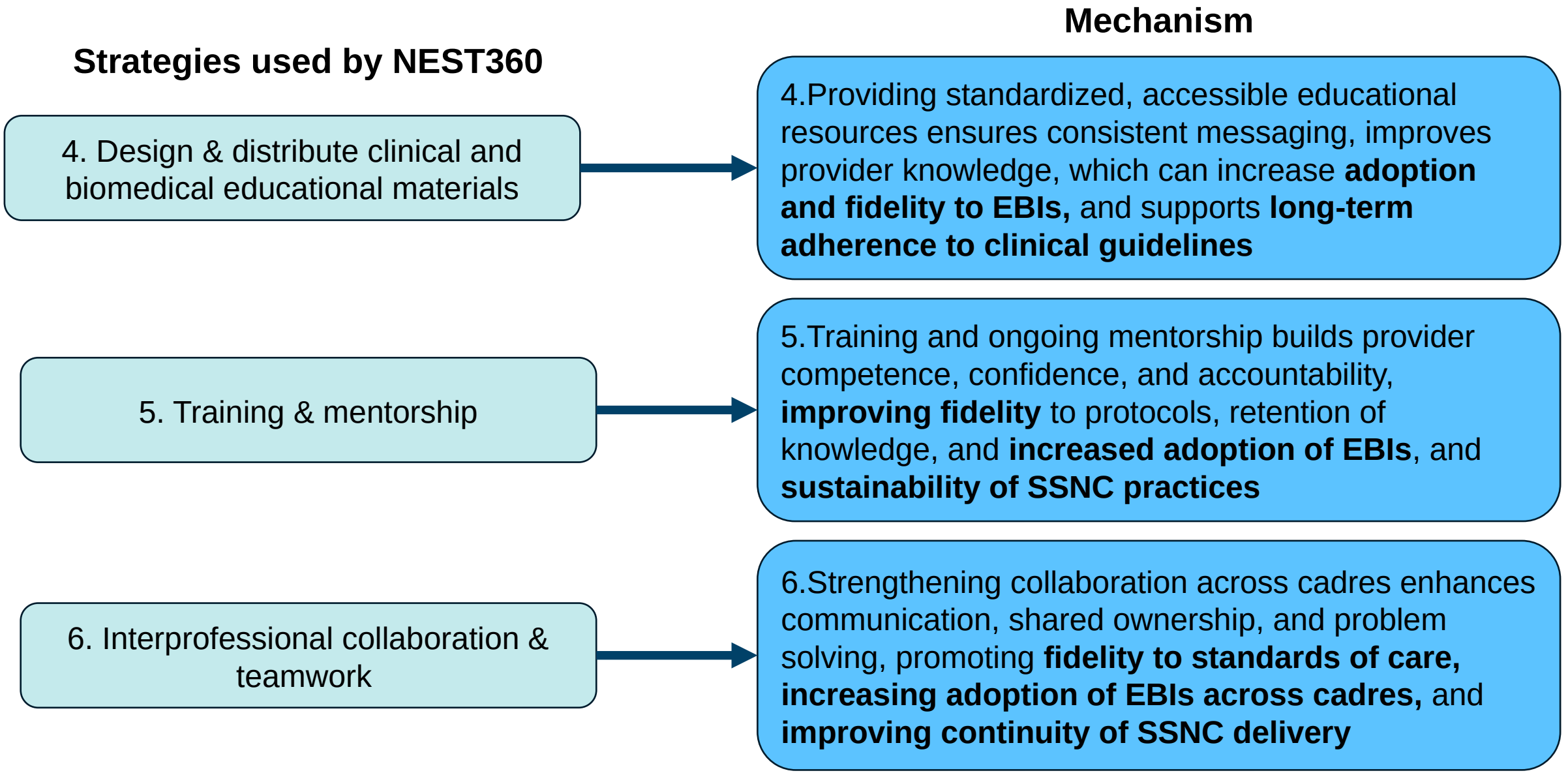
1. Engage policy makers & leadership
2. Quality improvement (QI) activities
3. Develop data systems to capture & feedback data and support routine SSNC data collection
4. Design & distribute clinical and biomedical educational materials
5. Training & mentorship
6. Interprofessional collaboration & teamwork
7. Motivate staff and empower collaboration
8. Ensuring the availability of appropriate functional SSNC medial devices
9. Identify & qualify appropriate technology
10. Hatch Technologies (African-based technology distribution)
11. Engage family members in a care process and follow up actions
12. Ensure adequate funding
13. Tailoring local infrastructure

Colors coordinates to the tier 1 model

ategies & Mechanisms Part 1

Strategies used by NEST360	Mechanism
1. Engage policy makers & leadership	**1.** Policy-maker and leadership engagement fosters alignment of SSNC with national health priorities, **promotes long-term commitment,** and ensures **integration of SSNC into broader health system goals**
2. Quality improvement (QI) activities	2.Routine data-driven QI cycles build a **culture of learning and accountability,** enabling timely identification of gaps and driving **sustainable improvements in SSNC coverage and quality**
3. Develop data systems to capture & feedback data and support routine SSNC data collection	3.Robust, context-appropriate data systems enable **real-time monitoring**, feedback, and course correction, **supporting transparency,** and **improving responsiveness to local needs**

Tier 2: Strategies & Mechanisms Part 2



Tier 2: Strategies and Mechanisms Part 3

Strategies used by NEST360	Mechanism
7. Motivate staff and empower collaboration	7.Recognition, supportive supervision, and empowerment of staff **increases acceptability, motivation, and engagement** leading to greater accountability, reduced burnout, and **higher quality care delivery**
8. Ensuring the availability of appropriate functional SSNC medical devices	8.Ensuring appropriate, functional technology is available at the right level of care enables providers to delivery life-saving interventions safely and effectively, promoting **feasibility and adoption of EBIs**, and **reducing gaps in SSNC delivery**
9. Identify & qualify appropriate technology	9.Identifying, testing, and sharing results of devices that can function in rugged settings **increases the feasibility of their use to provide SSNC preventing device graveyards** and **increasing the adoption of SSNC EBIs** because devices remain functional and available

Tier 2: Strategies and Mechanisms Part 4

Strategies used by NEST360	Mechanism
10. Hatch Technologies (Africa-based technology distribution)	10.Reliable, context-appropriate technology distribution channels improve **equitable access to devices, reduce downtime** from broken or missing equipment, and promotes **sustainability of SSNC infrastructure in low-resource settings**
11. Engage family members in a care process and follow up actions	11.Actively involving families in care builds trust, **increases caregiver competence and adherence to recommended practices,** and **promotes continuity of care after discharge,** improving neonatal survival
12. Ensure adequate funding	12.Securing and sustaining funding streams enables consistent procurement of supplies, **maintenance of infrastructure,** and **retention of staff capacity,** ensuring **sustainability of SSNC services**
13. Tailoring local infrastructure	13.Tailoring local infrastructure creates **safer, more efficient care environments, improving feasibility, quality,** and **sustainability of SSNC services**

Tier 2: Outcomes

- **Linkage to maternal care**
  - Global-level
  - National-level
  - Hospital-level
- **Family centered care**
  - Global-level
  - National-level
  - Hospital-level
- **Post-discharge follow-up**
  - Global-level
  - National-level
  - Hospital-level
- **Quality improvement**
  - Global-level
  - National-level
  - Hospital-level
- **Equipment and commodities**
  - Global-level
  - National-level
  - Hospital-level
- **Referral process**
  - Global-level
  - National-level
  - Hospital-level
- **Infrastructure**
  - Global-level
  - National-level
  - Hospital-level
- **Financing**
  - Global-level
  - National-level
  - Hospital-level
- **Impact**
  - Coverage rates
  - Mortality change
- **Leadership and governance (examples)**
  - National-level: SSNC and biomed allocations in annual budget
  - Hospital-level:
- **Data systems (examples)**
  - Global-level: # of MOH NEST-IT logins
  - National-level: % of countries logging into NEST-IT monthly
  - Hospital-level: Data entry delays and data validity
- **Human resources (examples)**
  - Global-level: # of clinicians that have receive NEST SSNC training
  - National-level: % of hospitals with SSNC trained staff
  - Hospital-level: Trained biomed at the hospital (Y/N)

Tier 3

This is where we report what NEST did and give practical resources for what others can do

Implementation factors are labeled as a barrier or facilitator based on NEST's experience, linked to strategies, and we explore the level of control we had over each factor

Tier 3

This is where we report what NEST did and give practical resources for what others can do

Discrete implementation strategies are listed under the general group of strategies

Tier 3

This is where we report what NEST did and give practical resources for what others can do

Each strategy group has a description for how those activities impacted the global, country, and hospital level

TIER 3 GRAPHIC

This is where we report what NEST did and give practical resources for what others can do

We provide a detailed explanation of the outcomes that NEST360 tracked including numerator, denominator, definition, and frequency of measurement

Tier 3: Implementation Factors Part 1

- **Data Systems**
  - Fragmented neonatal health forms, registers, and data systems at facility level capturing SSNC sensitive indicators ↓ (3)
  - No SSNC sensitive indicators at facility and national levels ↓ (3)
  - Poor data access and utilization ↓ (1,2,3)
  - Poor data quality ↓ (2,5)
  - Poor data management and governance↓ (1,3) **or is this in influence and in control?**
  - Lack of dedicated neonatal data clerk ↓ (1,12) **or is this in influence and in control?**
- **Human Resources**
  - Poor health workers attitudes/interest in SSNC ↓ (1,5,6,7)
  - Poor health workers competence/confidence in SSNC ↓ (1,5,6,7)
  - Inconsistent preservice training in SSNC ↓ (1)
  - Training available for SSNC ↑/ ↓ (5)
  - Inconsistent BMET training ↓ (5)
  - HC Provider awareness /knowledge ↑ / ↓ (2,4,5,6)
  - Supporting staff not following IPC guidelines (janitors, guard, support runners) ↓ (2,5,6)
  - BMETs unavailable ↓ (3,5)
  - Inadequate number of health workers ↓ (3,5) **or is this out of influence and out of control?**
  - Rotating staff and staff turnover ↓ (1) **or is this out of influence and out of control?**

Up arrow- Facilitator
Down arrow- Barrier
Number- Corresponds to a group of strategies
Green- In sphere of influence & control
Blue- In sphere of influence, outside sphere of control
Purple- Outside sphere of influence & control

Tier 3: Implementation Factors Part 2

- **Quality Improvement**
  - Inconsistent QI knowledge at national, regional, and facility levels ↓ (1,2,4,5)
  - Absent or inactive hospital and ward improvement teams ↓ (1,2,4,5,6)
  - Inconsistent QI training ↓ (1,4)
  - Inconsistent availability of QI documentation tools ↓ (1,2,3)
  - Gaps in documentation completeness ↓ (1, 2,3)
  - Absent or inconsistent auditing of care quality (ex. MPDSA) ↓ (2,3,4,5)
  - National and regional QI structures and resources showed variability in their inclusion of SSNC-focused guidance ↓ (1,3)
- **Equipment & Commodities**
  - Unavailability of functional equipment and supplies at facilities ↓ (5,8,9,10)
  - No strong procurement system ↓ (10)
  - Technology not fit for purpose ↓ (9)
  - Skill labs/skill stations unavailable or underused ↓ (5,6,7)
  - Expensive technology ↓ (8,10,12)
- **Referral process**
  - Poor adherence to referral guidelines ↓ (5,6,2)
  - Poor pre and during referral communication and consultations ↓ (1,11)
  - Poor referral feedback mechanism ↓ (1,11)
  - Lack of appropriate transportation for referrals ↓ (1)
- **Linkage to Maternal Care**
  - Policies that don't facilitate/enable zero separate↓ (1,11)
  - Siloed antenatal, delivery, postnatal, and SSNC↓ (1,11)
  - Women leaving early due to competing demands at home ↓ (1,4,11)
- **Family centered care**
  - Lack of training and policies to support parental involvement ↓ (11)
  - Poor education/counseling for caregivers ↓ (4,6,11)
  - Family members lack awareness in handwashing and IPC ↓ (4,5,11)
  - Family interest in being involved in the care process ↑ / ↓ (4,11)
  - Lack of family involvement in decision making process ↓ (6,7,11)
  - Unavailability of bereavement care ↓ (1,4)
- **Post-discharge follow up**
  - No/poor discharge follow up and early child development ↓ (1)
  - No/poor follow up feedback mechanisms ↓ (1,3)
  - No/poor community health workers engagement ↓ (5,6,7)
  - Community based support and continuous care ↓ (1,4,11)

Tier 3: Implementation Factors Part 3

- **Infrastructure**
  - No standard guide for space for Level 2 SSNC ↓ (1,4,13) **or is this in influence and out control?**
  - Inadequate space for SSNC including KMC/immediate KMC ↓ (1,3,12,13)
  - Poorly designed space - leads to poor flow and poor IPC practices ↓ (1,12,13)
  - Poor flow of care of newborns from labor ward to NBU ↓ (1,13)
  - Unreliable electric power and poor backup power sources ↓ (1,12,13)
  - Poor medical gas and vacuum systems ↓ (1,8,13)
  - Insufficient WASH facilities↓ (1,12,13)
  - Blood culture are unavailable ↓ (1,8,9,12)
- **Financing**
  - Donor funding ↑ (6,12)
  - Government financing ↑ / ↓ (12)
  - Long-term funding ↑ / ↓ (6,12)
  - No budget allocation to SSNC in national and facility budgets ↓ (1,12)
  - Poverty ↓ (1)
  - Little insurance coverage ↓ (1)
  - Poor exempted service reimbursement system ↓ (1)

Implementation factors requests *(See poster)*

- Do we need to add any implementation factors?
- Do we need to remove any implementation factors?
- Do we need to modify any implementation factors?
- Do you agree with the barrier & facilitator labels for the implementation factors?
- Do you agree with the implementation factor color coding?

Provide your feedback on level of influence & orientation with this google survey

Feel free to add a post-it note to the poster with your answer(s)

Tier 3 strategies & mechanisms

Strategies

1. Engage policy makers & leadership
2. QI activities
3. Develop data systems to capture & feedback data and support routine SSNC fata collection
4. Design & distribute clinical& biomedical educational materials
5. Training & mentorship
6. Interprofessional collaboration & teamwork
7. Motivate staff & empower collaboration
8. Ensuring the availability of appropriate functional SSNC medical devices
9. Identify & qualify appropriate technology
10. Hatch technologies
11. Engage family members in care process and follow up actions
12. Ensure adequate funding
13. Tailoring local infrastructure



Mechanisms

- At the <u>global/program level</u> **[strategy]** entails XYZ, NEST performed these activities because XYZ can lead to 123.
- At the <u>country level</u> **[strategy]** entails XYZ, NEST performed these activities because XYZ can lead to 123.
- At the <u>hospital level</u> **[strategy]** entails XYZ, NEST performed these activities because XYZ can lead to 123.

Strategy requests *(See poster)*

- Do we need to add any new strategies?
- Do we need to remove any strategies?
- Do we need to modify any strategies?
- What NEST resources should we link at this level?

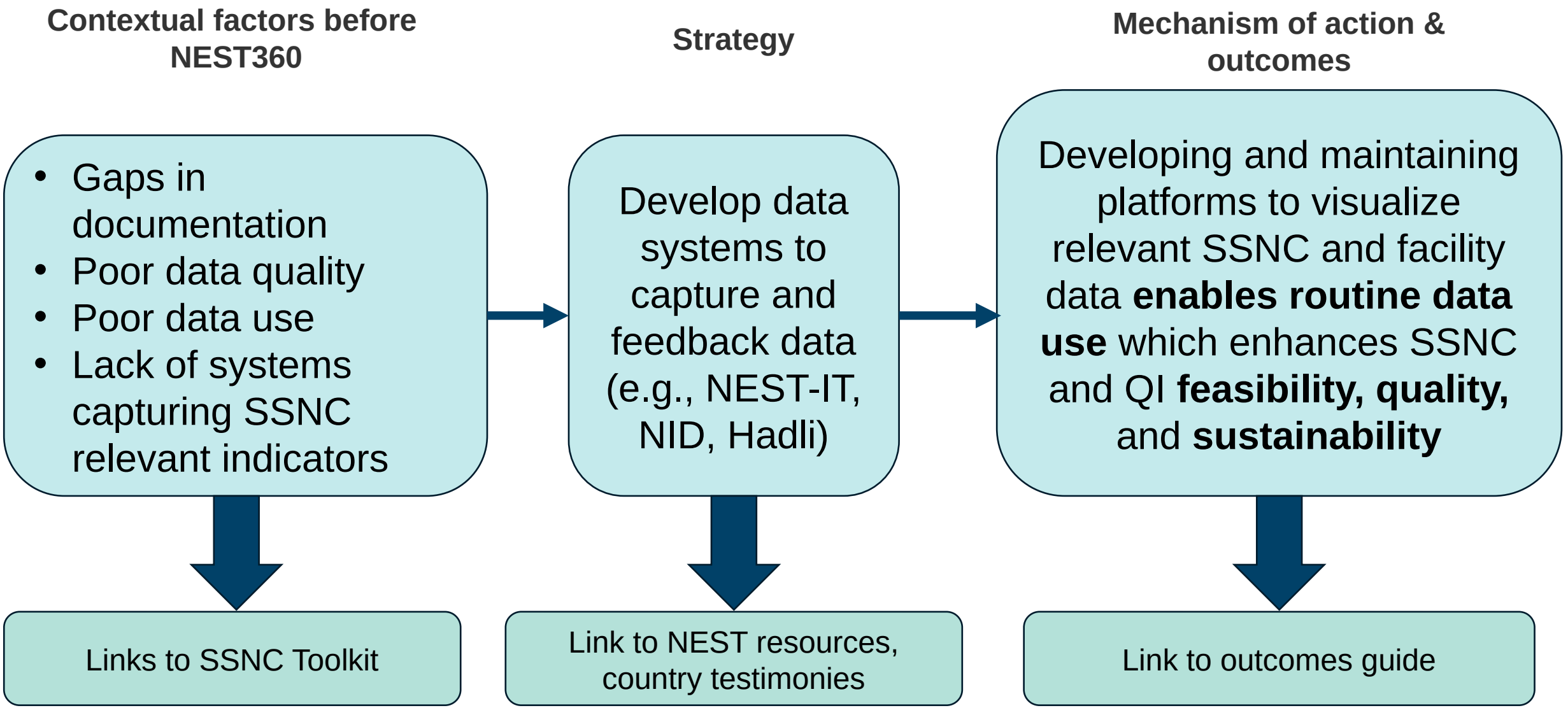
Feel free to add a post-it note to the poster with your answer(s)

Tier 3: Outcomes

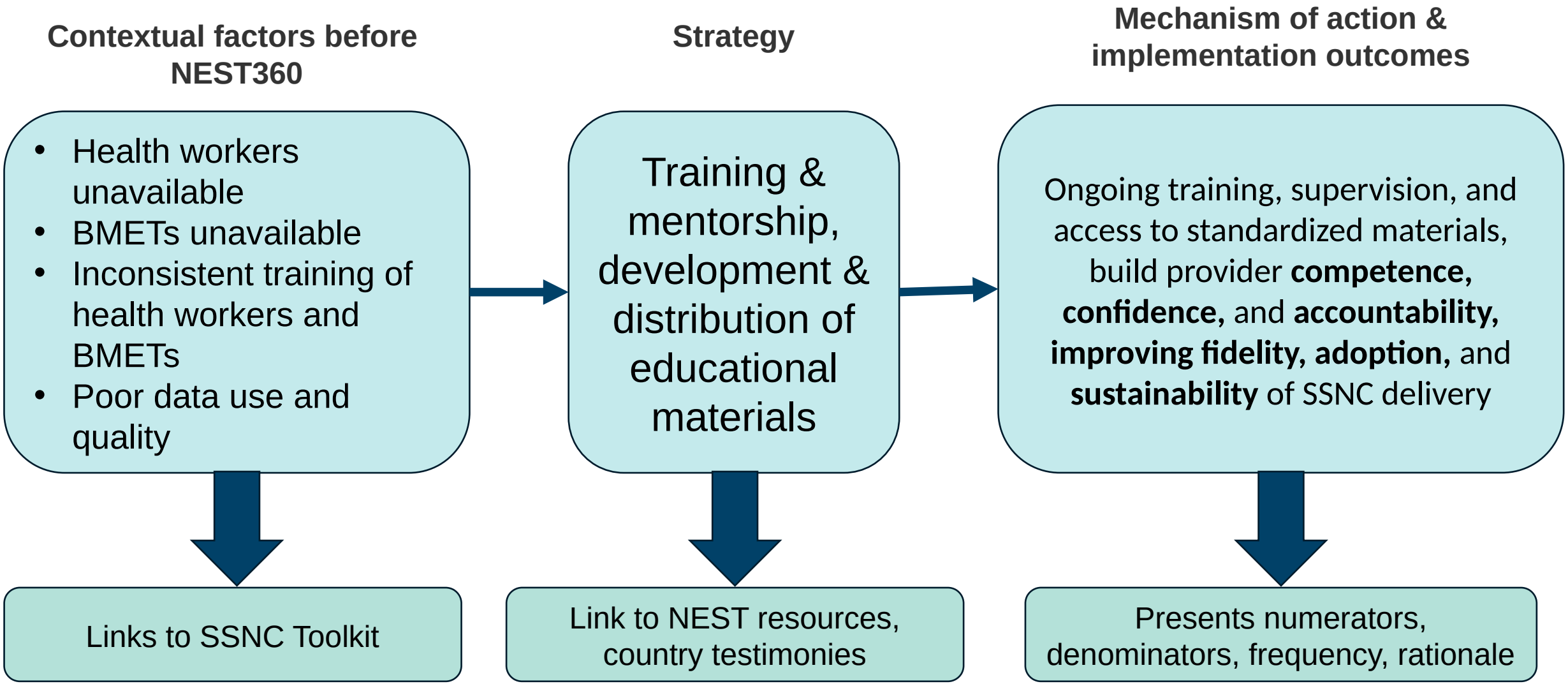
- Currently living in an excel sheet
- Outcomes from FCC report and NEST-IT

- Information included
  - Domain
  - Outcome
  - Definition
  - Rationale
  - Frequency of measurement
  - Numerator & denominator
    - Global level
    - Country level
    - Facility level

Walk through example



Walk through example



Questions?